Vibe Theory

A Simulable Discrete Universe:
How Space, Time, Matter, and Force Can Emerge from One Mesh

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Abstract

This paper tests Vibe Theory in simulation and assembles the results into a single picture. The framework holds that reality is one substance, the *vibe*, existing only as a mesh of relations (*notes*) among states (*tones*), with no background space or time, and that everything else is a large-scale pattern of that mesh. We built a finite, seeded, reproducible testbed and posed nine load-bearing questions, then measured the answers. Geometry is recovered sharply and almost uniquely from a discrete causal order, and, with a correct uniform-measure sampler, the substrate makes smooth low-dimensional spacetime a *stable phase* that the action supports, coexisting with the layered phase in the manner of a first-order transition, where without the action the smooth order decays. Whether that phase strictly dominates the sum over histories is left open. Matter is built from the same tones, with spin appearing as a topological invariant of the mesh and a single chiral fermion realized through the overlap construction. Force lives on the links as a gauge connection that confines, carries a topological index equal to the gauge charge, and forms a chiral condensate in dynamical Abelian and non-Abelian fields. The gravitational action is the discrete Benincasa-Dowker functional whose continuum limit is curvature, and it is exactly the term that makes spacetime stable. A hyperbolic random mesh is shown to be Lorentz-safe, navigable without addressing, and stable under growth. Quantum violation is shown to be reachable from the deterministic mesh, with the precise residual tension made quantitative. We give the full epistemic scoreboard, separating what is demonstrated, what has strong evidence but no proof, and what is genuinely open or beyond reach. The conclusion is that monism is computationally coherent and that the dynamics supports a stable spacetime phase, with the strict-dominance, quantum, and chiral-gauge questions stated precisely and left open. This is a simulable model with a concrete build target, not a finished empirical theory.

1. Introduction

This is the third presentation of Vibe Theory. The first (Pollard, 2025) set out the framework as philosophy. The second (Pollard, 2026) gave it a discrete mathematical structure built entirely from experience, with a concrete simulable target. This paper takes that target and runs it. It reports what a finite, reproducible simulation of the model actually shows, and it assembles those results into one picture of how space, time, matter, force, gravity, and quantum behaviour fit together in a universe made of one substance.

The central claim of Vibe Theory is monism. There is one kind of thing, the vibe, a state that stands in relation to other states. There is no background space and no background time. The only primitive relation is the note. Everything else, geometry and matter and force, is a large-scale pattern of the vibe mesh. A monist theory of this strictness owes a debt that most ontologies never pay. It must show that the variety of the physical world can actually come out of one substrate, not merely assert that it could. This paper is an attempt to pay that debt in simulation.

The method is deliberately empirical about the model, even though the model itself is not yet empirically tested against nature. We treat each foundational question as a measurement. We built a testbed, a finite and seeded program with a known-answer test suite, and posed nine questions that the framework must answer to be coherent: whether the dynamics can have a stable ground state and a local time, whether a smooth spacetime is dynamically stabilized rather than assumed, whether a substrate can be at once Lorentz-safe and navigable, whether matter and spin are patterns of the mesh, whether the recovered geometry is unique, whether the sum over histories is computable and lands on the right dimension, whether quantum statistics can come from a deterministic base, whether gauge force and a charged fermion can be built, and where the framework's own boundary lies. Each question becomes a runnable experiment, and each experiment returns a number.

The results are mixed in the honest way a real research program is mixed, and we present them that way. Several questions are answered cleanly and validated against known-answer cases. Two of the hardest, the dynamical stabilization of a spacetime phase and the resolution of the law, are addressed at the scale we can reach, the first showing a stable phase and a first-order signature with strict dominance still open, the second by relocating the Hamiltonian from the microscopic rule to the emergent mesh. One, the quantum link, is made precise rather than closed. A few are genuinely open in physics itself, or lie at the framework's philosophical boundary, and we mark them as such rather than overclaim.

Throughout, the framework sits in the family of discrete-physics and digital-physics programs. The substrate is a causal set in the sense of Bombelli, Lee, Meyer, and Sorkin (1987), the dynamics borrows the Benincasa-Dowker action (2010), the matter sector uses lattice fermions and the overlap construction of Neuberger (1998) and Ginsparg and Wilson (1982), the gauge sector uses Wilson's lattice gauge theory (1974), and the quantum question engages 't Hooft's Cellular Automaton Interpretation (2016) and Bell's theorem (1964). What Vibe Theory adds is the insistence that all of these live on one experiential substrate, and the demonstration, here, that they can.

This work is a philosophical and mathematical model with a simulable build target rather than a finished scientific theory. References to physics are sources of structure and tests of coherence, not claims to have derived nature. What the paper offers is a candidate ontology that has been put under simulation, with its successes, its partial results, and its open frontiers all stated precisely.

2. The Model in Brief

We recall only what is needed to read the results. The full derivation is in Pollard (2026).

Definition 1 (Vibe, tone, note, mesh) A *vibe* is the one kind of entity, a state in relation. Each vibe carries a finite *tone*, with the working alphabet ternary, $t \in \{-1, 0, +1\}$, and richer spinor tones reserved for matter. The *note* is the only primitive relation, v noting w meaning v is present to w. The *vibe mesh* is the whole population of vibes, their tones, and the note relation among them. There is no space or time outside the mesh.

The mesh grows and updates by a local *rule*, one *beat* at a time, each vibe feeling only its neighbors. Four committed choices shape it, each with named alternatives held open:

- **Discreteness.** The base is finite. Real numbers appear only as emergent large-scale measurements, never in the substrate.
- **Hyperbolic geometry.** The mesh branches outward exponentially rather than as a flat grid.
- **Eternal expansion.** The mesh keeps growing without bound.
- **No hidden state.** Everything is in the tones and notes. There is no separate bookkeeping behind them.

The causal order of the mesh, the relation of one vibe being in another's past, is a locally finite partial order, which is exactly a *causal set* (Bombelli et al., 1987). This identification is what lets the geometry, dynamics, and matter of the mesh be studied with the established discrete-spacetime toolkit, and it is the bridge between Vibe Theory's ontology and the rest of this paper.

The guiding picture for the results is a ladder. At the bottom is the mesh of tones and notes. Above it, as large-scale patterns, are space, time, matter, force, gravity, and the correlations we call quantum. The paper climbs this ladder one rung at a time, and the final section shows how the rungs connect into a single structure.

3. The Testbed

The model is fully discrete, so it is fully simulable. We built a testbed, an open-source TypeScript library, the `@cluesurf/vibe` package [1], available under the MIT License at `https://github.com/cluesurf/vibe`, that generates a discrete substrate, runs a local rule over it in discrete beats, and measures the emergent physics. Everything is finite and seeded, so every number reported here is a deterministic function of a stated seed and reproduces exactly.

3.1. Design

The library is organized as a small set of layers. *Substrates* generate the mesh: Poisson-sprinkled Minkowski and curved spacetime, regular lattices, hyperbolic tilings, hyperbolic random graphs, and classical sequential growth. *Rules* update it: synchronous, asynchronous, reversible, rewriting, and gauge updates. *Operators* act on it: the graph Laplacian, the Kahler-Dirac operator, the evolution Hamiltonian, the gauge-covariant Dirac operator, and the overlap operator. *Measures* read off physics: dimension, distance, curvature, manifold-likeness, Lorentz isotropy, navigability, the CHSH correlation, locality, integration, Wilson loops, and the chiral condensate. *Dynamics* provide the samplers: the Benincasa-Dowker action, causal-set Monte Carlo, parallel tempering, exact enumeration, a correct uniform-measure sampler, and the Wilson heat bath.

Real numbers appear only as measured outputs, such as sprinkling coordinates and eigenvalues, never as the substrate. This keeps the discreteness commitment honest at the level of the code.

3.2. Trustworthiness

A model is only as trustworthy as its checks. The testbed carries a suite of known-answer tests that must pass before any result is believed: a Poisson sprinkling of a d-dimensional region must recover dimension d, a regular lattice must be more anisotropic than a sprinkling, a CHSH experiment with independent settings must respect the classical bound, the overlap operator must have one species with exact chiral symmetry, the uniform-measure sampler must reproduce exact enumeration, and so on. Several early results were wrong in plausible-looking ways and were caught only by these checks, including a miscalibrated dimension estimator, a sprinkler that was not uniform by volume, an anisotropy statistic that cancelled for symmetric lattices, and a Monte Carlo move that did not sample the uniform measure. Each was fixed and recorded. The results below are reported only for measures that pass their known-answer tests.

3.3. The nine questions

We pose the model's load-bearing questions as P1 through P9, each a runnable experiment:

- **P1.** Can the dynamics have a bounded-below and local Hamiltonian (a stable ground state and a local time)?
- **P2.** Is there a dynamics under which manifold-like spacetime dominates?
- **P3.** Can one substrate be at once Lorentz-safe and navigable?
- **P4.** Is matter, including spin, a pattern of the mesh?
- **P5.** Is the geometry recovered from a discrete order unique?
- **P6.** Is the 2D sum over histories computable and 2-dimensional?
- **P7.** Can quantum statistics come from the deterministic mesh?
- **P8.** Can a gauge force and a charged fermion be built?
- **P9.** What is the relation between structure and experience?

The next sections answer these in the order of the emergence ladder, grouping them by the structure they build.

4. Space: Geometry from the Order, and Spacetime as a Phase

Space in Vibe Theory is not a background. It is the large-scale shape of the note structure. Two questions decide whether this works. First, does a discrete order actually carry a definite geometry (P5)? Second, is that smooth geometry forced by the dynamics, or merely assumed (P2, P6)?

4.1. The geometry is unique (P5)

A causal set fixes its continuum geometry through the Hauptvermutung conjecture: order plus number recovers a Lorentzian manifold essentially uniquely. We tested the sharpness directly. Eight independent Poisson sprinklings of the same Minkowski region each had their dimension recovered by the Myrheim-Meyer estimator.

Result 1 (Sharp recovered geometry) The recovered dimension across independent sprinklings of a fixed region is 3.02 ± 0.05, and the proper-time (longest-chain) distance across the region has coefficient of variation 0.027. The geometry a discrete order carries is sharp, not just in dimension but in distance.

So the smooth shape is genuinely latent in the discrete order. Continuous geometry is what the discrete vibe order looks like from far away, the way a beach looks smooth until the grains appear.

4.2. Spacetime as a stable phase (P2)

The harder question is why the order is manifold-like at all. Most causal sets are not. The Kleitman-Rothschild orders, flat three-layer orders, dominate the uniform measure at large size by sheer number. A theory that wants spacetime to be real must show that its dynamics suppresses these and makes smooth orders dominate. This is the deepest problem in causal set theory, open in the literature, and we treat it carefully.

The order parameter is the *height ratio*, the longest chain divided by $\sqrt{N}$. A 2D manifold-like order has height about $\sqrt{N}$ (its proper time), so the ratio is order one. A layered order has height three, so the ratio falls toward zero.

The first finding is negative and clarifying. The sharp Benincasa-Dowker action fails: minimised, it drives the ensemble to layered orders (height ratio 0.29, apparent dimension 3.45). The fix is the *smeared* action, a nonlocal kernel that averages over about $1/\epsilon$ order-layers, which tames the action fluctuations. The smeared action drives the ensemble toward manifold-like orders rather than layered ones, exactly confirmed by exact enumeration at small size (where the entropic competition has not yet set in).

But a subtlety nearly hid the real answer. The naive Monte Carlo move did not sample the uniform measure, so it could not even pose the dominance question. We built a correct uniform-measure sampler: propose toggling a single relation, accept only if the result is still a transitive order. That proposal is symmetric, so it samples the uniform measure exactly, validated against exact enumeration.

Result 2 (The sampler is correct) The uniform-measure sampler reproduces exact enumeration: at inverse temperature zero its manifold fraction is 72% at $N = 6$ and 84% at $N = 7$, matching the exact values to the digit. It then reaches the large-N entropic regime invisible at small size: the manifold fraction falls $92\%, 23\%, 4\%, 0\%$ at $N = 8, 16, 32, 64$, so the layered orders take over the uniform measure exactly as Kleitman-Rothschild predict.

With the entropic regime in hand, the dominance question can finally be asked, and the answer is a phase transition.

Result 3 (Coexistence of two phases, the first-order signature) At $N = 128$, under the smeared action, two long-lived phases coexist. From a cold start the order stays layered (manifold fraction 0%). From a warm start (a 2D sprinkling) it stays manifold-like (manifold fraction 100%, height ratio 1.2 to 1.6) for all

$\beta \geq 1$, whereas at $\beta = 0$ even a manifold start decays. Two long-lived phases at one coupling, separated by a barrier, is the signature of a first-order transition. The coexistence holds across $N = 48, 96, 160$, finite-size evidence that it persists toward larger systems.

So smooth spacetime is not merely assumed and trivially preserved. The substrate's own action makes it a stable phase, one that decays without the action and persists with it, coexisting with the layered phase in the way supercooled water and ice can both persist near freezing. What the testbed has not located is the freezing point itself, the coupling at which one phase becomes the lower-free-energy equilibrium. So we show that a stable spacetime phase exists and is supported by the dynamics, not yet that it strictly wins.

4.3. The 2D path integral lands on 2D orders (P6)

P6 is the 2D specialization, and it asks the sharper question of whether the stable phase is genuinely two-dimensional, not merely non-layered.

Result 4 (The stable phase is 2D) The warm-started stable manifold phase has Myrheim-Meyer dimension $2.03, 2.09, 2.05$ at $N = 64, 96, 128$. The exact enumeration agrees, with a dimension sweet spot near two at moderate coupling. The 2D sum over histories, sampled with the correct measure, lands on 2-dimensional manifold-like orders.

What remains open is which phase has the lower free energy, and so strictly dominates, at a given coupling. That is a free-energy comparison across the barrier, reachable by thermodynamic integration or tempering, and the continuum-limit proof is a separate mathematical question. There is also a residual caution at finite size: two long-lived phases under a finite chain are strong evidence of genuine metastable basins, but slow mixing cannot be fully excluded without the free-energy crossing. What is done is the harder existence half, that a stable spacetime phase exists at all and is supported by the dynamics where without it the smooth order decays.

5. Time and the Law (P1)

Time in Vibe Theory is the depth of the note order, and the law is the rule that updates the mesh. P1 asks whether that law can have a stable ground state (a bounded-below energy) and a local notion of time (a local Hamiltonian) at once. A reversible local rule has a global update U, and its Hamiltonian is $H = i \log U$. The question is whether H is both bounded below and local.

5.1. The trilemma

We built H exactly from the cycle structure of U and expanded it in the Pauli basis, measuring how its operator weight is distributed over interaction range. A control validates the measure: a single-cell flip has $H = (\pi/2)(I - X_0)$, exactly range one, and the measure returns locality length 1.00. The reversible rules tell a sharp story.

Result 5 (The principal Hamiltonian is nonlocal) For every nontrivial reversible rule tested, the principal (minimal-energy) Hamiltonian has interaction range that grows with the system, with locality length about $0.8N$ for the XOR-parity rule at $N = 6, 8$. The bounded-below vacuum is easy, but the minimal-energy law is not local.

An explicit local branch exists for special rules. For a product of commuting local gates, $H = \sum(\pi/2)(I - G_{\text{block}})$ is a genuine logarithm of U, with bounded locality length and a bounded-below spectrum. But this requires the gates to commute, which means the rule does not propagate information. Putting these together yields a clean obstruction.

Principle 1 (The law is a trilemma) For a reversible cellular automaton's own Hamiltonian, local, bounded below, and information-propagating appear unable to hold all three at once. Any two are achievable. All three resist.

This is an obstruction observed across the rules we tested, at small size, and argued from the branch structure of the logarithm. It is a principle drawn from the evidence, not a theorem proved for all rules, and we state it as such.

5.2. The resolution

The way out is to stop demanding that the quantum Hamiltonian be the logarithm of one microscopic step. Define it instead as a local operator on the *emergent* mesh, the graph Laplacian or the Dirac operator.

Result 6 (The emergent Hamiltonian beats the trilemma) The graph Laplacian on the emergent mesh is local (interaction range one, independent of size), bounded below (spectrum starting at zero), and propagating (a localized state spreads at a finite speed, a lightcone) all at once. On a ring of $N = 24$ and $N = 48$ the range stays one and the spectrum stays in $[0, 4]$, and the mean spread grows linearly in time.

The reading for Vibe Theory is that the rule and the flow of time are different objects. The rule builds the geometry (Section 4). Time is what local operators do on the grown mesh. The trilemma is an obstruction for the log of a microscopic step, not for physics, because physical time lives on the mesh that the rule produces, not inside the rule.

6. Matter: Spin from Topology, and a Chiral Fermion (P4)

Vibe Theory builds matter from the same tones as everything else, with no new primitive. A particle is a persistent pattern of the mesh, and its spin is meant to come from the shape of the connections, not from an added ingredient. P4 tests this for the fermion.

6.1. The monist spinor and spin from topology

We build the Kahler-Dirac operator $D = d + \delta$ on the cochain spaces of a triangulated mesh. Its zero modes are the harmonic forms, whose count is a topological invariant, the sum of the Betti numbers. We built surfaces of known topology and counted.

Result 7 (Spin from topology) The Kahler-Dirac zero-mode count equals the surface Betti sum exactly: a disk gives 1, a cylinder gives 2, a torus gives 4. Every component of each zero mode is a cell tone, so the spinor is built entirely from the substrate. The statement that a tone is a small spinor is realized, and the spinor's zero modes are a topological invariant of the mesh, which is the spin-from-topology reading (in the spirit of Bombelli et al., 1987).

6.2. The chirality wall, threaded

The hardest matter fact is that a single chiral (handed) fermion cannot live on a naive lattice. The Nielsen-Ninomiya theorem forces doublers: a local, hermitian, chirally symmetric lattice Dirac operator comes with equal numbers of left- and right-handed copies. We measured this and its resolutions in 2D momentum space.

Result 8 (Overlap threads Nielsen-Ninomiya) The naive lattice operator has 4 species (the 2^2 doublers). The Wilson operator removes them to 1 but breaks chiral symmetry (a large Ginsparg-Wilson residual). The overlap operator $D = 1 + \gamma_5 \,\mathrm{sign}(H_W)$ has 1 species and an exact lattice chiral symmetry, with Ginsparg-Wilson residual 6×10^{-16}, machine zero. A single chiral fermion exists on the mesh, with exact chiral symmetry, threading the no-go theorem.

So matter is monist and its spin is topological, and the deepest obstruction to chiral matter is passed at the kinematic level. The remaining wall, a fully interacting Weyl-projected chiral gauge theory, is open in physics itself and is discussed with the forces.

7. Force: Connections on the Links (P8)

A force in Vibe Theory is an instruction carried along a note, the twist applied to a tone as it crosses a link. This is exactly a gauge connection. P8 climbs a ladder: a gauge field on the links (Stage A), coupled to a charged fermion (Stage B), confining (Stage C), and feeling matter and topology (Stage D and beyond).

7.1. Confinement

We put an SU(2) gauge field on a periodic lattice, with link variables as unit quaternions and the Wilson action, and measured the static potential through the Creutz ratio (the string tension).

Result 9 (Non-Abelian confinement) In 3D SU(2) lattice gauge theory the string tension is positive at every coupling and falls from 1.32 to 0.40 as β rises, while the average plaquette climbs from 0.10 to 0.65. A positive string tension is the Wilson area law, the hallmark of confinement, the defining behaviour of the strong force.

7.2. The fermion sees gauge topology

We coupled the chiral overlap fermion to a U(1) gauge background of known topological charge Q and computed its index from the spectral asymmetry of the kernel H_W, using a complex-Hermitian eigensolver built for the purpose.

Result 10 (The lattice index theorem) The overlap fermion's index equals the gauge topological charge exactly: index $= -Q$, an integer, for every charge tested, with the gauge flux equal to $2\pi Q$ and the kernel Hermiticity error zero. This is the lattice Atiyah-Singer index theorem. The chiral fermion sees gauge topology exactly, at finite lattice spacing.

7.3. The chiral condensate

Finally we coupled the overlap fermion to a *dynamical* gauge field, the Schwinger model and its non-Abelian cousin, and measured the chiral condensate through the Banks-Casher near-zero spectral density of the gamma-five-Hermitian overlap.

Result 11 (A condensate forms from the anomaly) In a dynamical Abelian gauge field the chiral condensate signal is zero in the free theory and nonzero once the field is on, rising at moderate coupling. The same pattern holds in a dynamical non-Abelian SU(2) field. The chiral fermion feels the fluctuating field and condenses, the anomaly-induced behaviour of the Schwinger model, in both the Abelian and the non-Abelian case.

So the link of Vibe Theory and the gauge connection of physics are the same object. Force is the instruction on the connection, it confines, it carries a topological index that ties it to matter, and it produces a condensate. Two honesty notes belong here. These are small-lattice results by exact diagonalisation: the string tension is the Creutz-ratio estimator rather than the asymptotic value, and the condensate is the Banks-Casher near-zero density (a signal proportional to the condensate) rather than its precise Schwinger value, which would need the dynamical fermion determinant and a continuum extrapolation. And the genuinely open rung, the Weyl-projected non-Abelian chiral gauge theory, has no satisfactory non-perturbative construction anywhere in physics, so we reach the rung below it (a vector-like fermion in the field) rather than the wall itself.

8. Gravity: the Action that Shapes the Mesh

Every other force lives on the mesh. Gravity is the mesh. It is the responsiveness of the geometry of relations to what the relations carry. In Vibe Theory this is not a separate result but a reading of the dynamics already in hand.

The action that drives the causal-set Monte Carlo is the Benincasa-Dowker functional (Benincasa and Dowker, 2010), the discrete analogue of the Einstein-Hilbert action, whose continuum limit is the Ricci scalar curvature integrated over spacetime. It is built from the abundances of small order intervals, and it is the discrete curvature of the note structure.

Principle 2 (Gravity is the stabilizing action) The same action whose smeared form makes smooth spacetime a stable phase (Section 4) is the discrete gravitational action. The dynamics that stabilizes a smooth geometry and the dynamics of gravity are one functional, not two.

This unifies two things that look separate. In Section 4 the smeared Benincasa-Dowker action was the term that stabilized the manifold phase against the entropic layered background. Here we name what that term is: it is gravity. So in Vibe Theory the stability of a smooth spacetime and the existence of a gravitational action are the same fact, seen once as a phase and once as a law of curvature.

What we have shown is the action and its phase, not the full field equations. We have not derived the Einstein equations as equations of motion, nor a graviton as a propagating excitation. Those are the natural next steps, and they are program rather than result. The standing claim is the one above: the functional that shapes the mesh into a smooth, curved geometry is the discrete Einstein-Hilbert action, and it is the same functional that makes spacetime stable.

9. The Substrate: Lorentz-Safe and Navigable at Once (P3)

Vibe Theory commits to a hyperbolic, expanding mesh with no hidden state. P3 is the question this most directly tests, and it is the most framework-specific result. The mesh must be two things that seem to pull against each other. It must respect relativity, having no preferred frame, which random structures give but rigid ones do not. And it must be navigable, so a vibe can route influence to a distant target using only what it locally knows, which rigid addressable structures give but generic random ones do not. Can one substrate be both?

9.1. The both-worlds substrate

We compared substrates on three measures at once: exponential reach (ball growth), Lorentz isotropy (an angular order parameter, near zero meaning no preferred frame), and navigability (greedy and backtracking geometric routing using only local neighbors and a coordinate).

Result 12 (All three at once) A connected hyperbolic random graph (mean degree about 11) has exponential reach, anisotropy 0.07 (as Lorentz-safe as a pure sprinkling), and 100% navigability with backtracking (greedy alone reaches 97%). A regular lattice fails (anisotropy 1.0, a hard preferred frame, and not navigable). A flat sprinkling is isotropic but neither reaches nor is navigable. The hyperbolic mesh is the substrate that is relativistic and navigable at once.

The price is giving up exact addressing for greedy geometric routing, which needs only a coordinate and the local neighbors, consistent with the no-hidden-state commitment. There is no global routing table.

9.2. Stable under growth

A one-shot construction is not a living mesh. We tested whether the property survives the eternal expansion the framework commits to, modelling growth as an expanding hyperbolic disc at constant density.

Result 13 (Survives expansion) Across an eightfold growth of the mesh at constant density, the both-worlds property persists at every size: exponential reach, anisotropy near 0.1, and 100% backtracking navigability hold throughout. The substrate stays relativistic and navigable as it grows.

This is the result where Vibe Theory's committed choices pay off most directly. The hyperbolic geometry is load-bearing, not stylistic, because only it gives reach, Lorentz safety, and navigation together. The expansion preserves the property. And navigation needs no hidden bookkeeping. The three commitments stand or fall as one package, and here they stand.

10. Quantum Behaviour from the Deterministic Mesh (P7)

The deepest question for any classical-base ontology is whether it can carry quantum statistics. Bell's theorem (1964) says no local model of independent coins can match the quantum CHSH correlations, unless one gives up the assumption that the measurement settings are statistically independent of the hidden state. A monist mesh with no hidden state denies exactly that assumption, because the settings are made of the same substrate as the hidden state. P7 makes this quantitative.

10.1. Determinism makes violation possible

We swept how strongly the measurement settings track the shared hidden state, in a constructed superdeterministic model.

Result 14 (The cost curve) As the setting-state correlation rises from independence to full sharing, the CHSH value climbs $1.0, 1.6, 2.3, 3.1, 4.0$, crossing the classical bound of 2 near half-sharing, passing the quantum Tsirelson bound of 2.83, and reaching the algebraic maximum of 4. The price of denying statistical independence is quantified: a given amount of measurement dependence buys a given amount of violation.

10.2. Aligned bits, not bits

But the dependence must be of the right kind. We measured the CHSH value alongside the measurement dependence (the mutual information between settings and hidden state, in bits), refining the bound of Hall (2010).

Result 15 (The currency is aligned bits) At one bit of measurement dependence, an aligned correlation (tracking the feature the outcomes use) reaches $S = 4$, while a misaligned correlation of the same one bit gives $S = 1$, and a correlation below the measurement resolution gives nothing. So measurement dependence is necessary but radically insufficient. One bit can buy anything from no violation to maximal violation, depending on alignment. The currency of Bell violation is aligned bits, not bits.

10.3. The residual tension, from dynamics

The honest frontier is whether a natural mesh produces the aligned correlation without fine-tuning. In a relativistic mesh the shared past of two spacelike measurements (the overlap of their backward light cones) shrinks with their separation.

Result 16 (Violation decays with separation) Modelling the shared-past fraction as $e^{-d/\xi}$ in separation d, the CHSH value decays from 4.0 at zero separation to the classical bound within a couple of correlation lengths ($2.58, 1.87, 1.04$ at increasing d). Quantum mechanics gives a separation-independent violation. So a natural causal mesh reproduces quantum violation only for nearby measurements, not spacelike ones, unless the correlation is fine-tuned not to decay.

This is the precise residual difficulty of the classical-base reading, now a number rather than a slogan. Vibe Theory makes Bell violation possible from a deterministic mesh, and pins the exact remaining puzzle: why the correlation is aligned and why it would be separation-independent. We made the tension quantitative. We did not resolve it, and we do not claim to. This is the framework's most contested and least settled rung, and it is the one to watch.

11. The Boundary: Structure and Experience (P9)

Vibe Theory is, at root, a theory of experience. The vibe is a unit of experience, and the axiom is that to exist is to experience or be experienced. P9 asks what the testbed can and cannot say about this.

What it can say is structural. We can locate, on the mesh, where a self would sit on the framework's own terms: a region that is well screened from its surroundings (a Markov blanket) and highly integrated within itself.

Result 17 (Structural correlates of a self) On a hyperbolic mesh with random tones the testbed measures a Markov-blanket screening score of 0.77 and an integration proxy of 1.91, locating a well-screened, well-integrated region, the structural signature the framework predicts a self would have.

What it cannot say is whether those correlates *are* experience. That a region is well screened and highly integrated is a fact about structure. Whether it feels like anything is the framework's foundational claim, not a measurement the simulator can make. This is the honest boundary, and it is by design.

The point is worth stating plainly, because it is where Vibe Theory differs from theories that try to derive consciousness. Vibe Theory does not derive experience. It posits experience as the foundation and derives the physical from it. So the testbed is the right tool for the physical ladder, geometry through force, and the wrong tool for the foundational claim. We measure the structure. The experiential reading is the framework's jurisdiction, and we leave it there rather than dress a structural number as a solution to the hard problem of consciousness (Chalmers, 1995).

12. The Final Picture: How the Pieces Fit

The results are not a list. They assemble into one structure, and this section draws it.

12.1. The structure in one image

There is one substance, the vibe, existing only as a web of notes among tones, with no space or time behind it. The web grows by a local rule, which is the discrete gravitational action (Section 8), and that action makes a smooth low-dimensional geometry a stable phase (Section 4). Seen from far away the web has a shape, and that shape is *space*. The depth of its causal chains is *time*, and the local operator that runs on the grown mesh, not the log of the microscopic rule, is the flow of time and energy (Section 5). Stubborn topological patterns of the tones are *particles*, and their spin is a topological invariant of the weave (Section 6). The twist instructions carried along the notes are the *forces*, which confine and tie themselves to matter through topology (Section 7). Because the whole web is one connected structure grown from a common root, distant parts share a family resemblance, which is the root of *quantum* correlation (Section 10). And the mesh that carries all of this is hyperbolic, expanding, and navigable without hidden state, the both-worlds substrate the framework committed to (Section 9).

Everything is one kind of thing, related to itself in different patterns. The rungs are not independent. The same hyperbolic, expanding commitment that makes the substrate navigable (P3) is the geometry whose smooth phase the gravitational action stabilizes (P2, gravity). The same topology that gives the fermion its spin (P4) is what the gauge field couples to through the index (P8). The same connectedness that makes the mesh one structure (P3) is what allows the quantum correlation (P7). The law that grows the mesh (P1, gravity) and the time that runs on it (P1, the Laplacian) are two faces of one dynamics. The picture is tight: pull one thread and the others move.

12.2. The dictionary

Reading it as a translation, standard physics on the left, the vibe object in the middle, the evidence on the right.

Standard physics	Vibe Theory	Shown by
spacetime manifold	the smooth shape of the note web	P5, P2, P6
proper time, causal order	depth of the before-after chains	P5, P1
energy, the Hamiltonian	local operator on the grown mesh	P1
fermion, spin, chirality	topological pattern of tones	P4
gauge force (EM, strong)	the twist carried by each note	P8
confinement	area law of the link variables	P8
gravity, curvature	the flexing of the web's geometry	P2, gravity
quantum entanglement, Bell	family resemblance in one mesh	P7
relativistic, navigable space	hyperbolic expanding mesh	P3
the laws of physics	one local growth rule on the mesh	P1, P2

12.3. What the picture buys

The hardest thing a monist theory must do is produce the variety of the world from one substance. The dictionary is the evidence that Vibe Theory does this, not by assertion but by construction, for geometry, time, matter, force, the onset of gravity, and the onset of quantum behaviour. And it does more than describe: the substrate's own dynamics makes a stable spacetime phase that would otherwise decay, so the smooth world is supported by the law rather than only assumed. That is movement from a coherent description toward a generative theory, for the physical ladder. The crossing is not complete, because we have shown the spacetime phase is stable and supported but not yet that it strictly dominates, but the direction is clear.

13. The Epistemic Scoreboard

A theory is only as good as its honesty about what it has and has not shown. This section is the ledger. It sorts every claim into what is demonstrated, what has strong evidence but not proof, and what is genuinely open or beyond the testbed.

13.1. What is demonstrated

These are validated against known-answer cases on the testbed.

- Geometry is recovered sharply and almost uniquely from a discrete order (P5).
- The substrate makes smooth 2D spacetime a stable phase that coexists with the layered phase (a first-order signature), with the correct sampler validated against exact enumeration (P2, P6). Whether it strictly dominates is in the next tier.
- Spin is a topological invariant of the mesh, and a single chiral fermion exists with exact lattice chiral symmetry (P4).
- Gauge force confines, carries a topological index equal to the gauge charge, and forms a chiral condensate in dynamical Abelian and non-Abelian fields (P8).

- A hyperbolic mesh is Lorentz-safe, navigable without addressing, and stable under growth (P3).
- The law is a trilemma, resolved by the emergent-mesh Hamiltonian (P1).
- Determinism makes Bell violation reachable, and its price is aligned bits (P7).

13.2. What has strong evidence, not proof

- That the manifold phase strictly *dominates* (not merely coexists) in the continuum. We show a sharp first-order coexistence holding across $N = 48$ to 160, but the free-energy crossing and the continuum-limit proof are not done.
- That a monist mesh carries quantum statistics. We show violation is reachable and quantify its price, but only an aligned, separation-independent correlation works, and we have not produced one from natural dynamics.

13.3. What is open or beyond reach

- Whether a *natural* mesh produces the aligned, separation-independent correlation without fine-tuning. The violation decays with separation in a natural mesh, while quantum violation does not. This is the contested heart of superdeterminism, unresolved here and in the literature.
- The Weyl-projected non-Abelian chiral gauge theory, open in lattice physics everywhere, not only here.
- The full Einstein equations and the graviton, beyond the gravitational action we exhibit.
- The specific Standard Model content: the gauge group, three generations, and the mass spectrum.
- Whether the mesh's structural correlates of a self *are* experience (P9). This is the framework's foundational posit, not a measurement the simulator can decide.

13.4. Reproducibility

Every numerical claim in this paper is produced by the open testbed [1] and is a deterministic function of a stated seed. The known-answer test suite passes (28 of 28 at the time of writing), the code typechecks cleanly, and each result has a corresponding experiment script and a written record. The numbers reproduce exactly.

14. Conclusion

This paper put Vibe Theory under simulation and assembled the results into one picture. From the framework's two commitments, that reality is one substance and that the substrate is finite, we built a reproducible testbed and asked nine load-bearing questions, then measured the answers.

The headline is twofold. Monism is computationally coherent: geometry, time, matter, force, and the onset of gravity and quantum behaviour all come out of one mesh of tones and notes, with nothing added, as the dictionary of Section 12 records. And the dynamics does more than describe it: the substrate's own action, the discrete gravitational action, makes a smooth low-dimensional spacetime a stable phase that decays without the action and persists with it, coexisting with the layered phase in a first-order signature. So the smooth world is supported by the law, not merely assumed. Whether that phase strictly dominates is the open refinement. This is movement from a coherent description toward a generative theory for the physical ladder, a direction rather than a completed crossing.

The picture is also tight. The hyperbolic, expanding, no-hidden-state commitments are not independent style choices. They reinforce one another: the geometry that makes the substrate navigable is the geometry whose smooth phase the gravitational action stabilizes, the topology that gives the fermion its spin is what the gauge field couples to, and the connectedness that makes the mesh one structure is what allows the quantum correlation. The commitments stand or fall as a package, and so far they stand.

We have been equally precise about the limits. Which phase strictly dominates in the continuum is strong evidence, not proof. The quantum link is made quantitative, not closed: a natural mesh's correlation decays with separation while quantum violation does not, and that tension is the framework's most contested rung. The Weyl-projected chiral gauge theory, the full field equations of gravity, the specific Standard Model content, and the foundational claim about experience are open in physics or lie at the framework's boundary, and we marked each as such rather than overclaim.

What Vibe Theory offers, then, is a candidate ontology that has been simulated rather than only argued. Its physical ladder is built and largely validated, its hardest dynamical question is advanced to a stable phase and a first-order signature at the scale we can reach with strict dominance still open, its quantum question is sharpened to a precise open problem, and its boundary is named honestly. The substrate is one kind of thing, related to itself in different patterns, and the testbed shows that a great deal of the world really does come out of it. The next steps are clear: the free-energy crossing, the natural alignment of the quantum correlation, and the climb past the chiral wall. They are stated precisely enough to be attempted, simulated, and criticized, which is the most a model at this stage can ask.

15. References

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